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Latency-Resilient Cooperative Multi-Agent
Reinforcement Learning for Adaptive Traffic
Signal Control

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Declaration of Candidate

We hereby declare that this thesis is our own original work and has not been submitted for any degree or diploma in any other university.

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Dedicated to our parents, whose love and support have been a constant source of strength.

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Abbreviations

AI	Artificial Intelligence
RL	Reinforcement Learning
MARL	Multi-Agent Reinforcement Learning
DQN	Deep Q-Network
DDQN	Double Deep Q-Network
SUMO	Simulation of Urban Mobility
MDP	Markov Decision Process
POMDP	Partially Observable Markov Decision Process
DEC-POMDP	Decentralized Partially Observable MDP
QoS	Quality of Service
V2I	Vehicle-to-Infrastructure
V2V	Vehicle-to-Vehicle
ITS	Intelligent Transportation System
RTT	Round Trip Time
API	Application Programming Interface
CTDE	Centralized Training with Decentralized Execution
MADDPG	Multi-Agent Deep Deterministic Policy Gradient
QMIX	Q-value Mixing Network
MAC	Multi-Agent Actor-Critic
GNN	Graph Neural Network
GCN	Graph Convolutional Network
AoI	Age of Information
IDM	Intelligent Driver Model
SAFE	Stochastic Automaton Follow-the-leader Engine

AIC	Akaike Information Criterion
BIC	Bayesian Information Criterion
TD	Temporal Difference

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Abstract

Urban traffic congestion remains one of the most significant challenges faced by modern cities, resulting in economic losses, environmental degradation, and reduced quality of life. Traditional traffic signal control systems rely on fixed or rule-based timing strategies that cannot adapt effectively to dynamic traffic patterns. Recent advances in Artificial Intelligence, particularly Reinforcement Learning (RL), have introduced promising approaches for adaptive traffic signal control. Among these, Multi-Agent Reinforcement Learning (MARL) allows multiple intersections to operate as intelligent agents that coordinate their decisions to optimize overall traffic flow.

Despite these advancements, most existing MARL-based traffic control systems assume ideal communication conditions between agents. In real-world smart city deployments, however, communication networks introduce delays, packet loss, and asynchronous updates that may significantly affect coordination between intersections. This gap between theoretical assumptions and practical deployment conditions remains a critical barrier to the real-world adoption of MARL-based traffic control systems.

This research proposes a latency-resilient cooperative MARL framework for adaptive traffic signal control. The proposed system systematically investigates the impact of communication latency and packet loss on cooperative traffic signal agents and explores theoretical mechanisms that allow the system to maintain stable performance under such network imperfections. The framework incorporates a latency-resilient mechanism that enables agents to utilize the most recent available information, estimate missing data using historical observations, and reduce dependency on synchronous communication.

The theoretical analysis examines system behavior under four distinct configurations: ideal communication (baseline), moderate imperfections, severe imperfections, and extreme imperfections. Performance is evaluated through established metrics including average travel time, queue length, waiting time, and throughput. A comprehensive simulation design is presented to guide future empirical validation. The theoretical findings provide insights into the robustness of MARL-based traffic control under realistic communication environments and contribute toward more deployable intelligent transportation systems.

Introduction

1 Background and Context

Urban traffic congestion constitutes a complex and multidimensional challenge that has been exacerbated by accelerated urbanization and the global increase in private vehicle ownership. The economic consequences of congestion are considerable, with empirical studies estimating annual losses amounting to billions of dollars as a result of diminished productivity, elevated fuel consumption, and environmental deterioration. Central to this issue is the traffic signal control system, which functions as the principal mechanism for regulating vehicular movement across urban road networks.

Conventional traffic signal control systems may be broadly classified into three generations. The first generation consists of fixed-time or pre-timed systems, in which signal timing plans are established offline on the basis of historical traffic data. Although these systems are relatively simple and reliable, they lack the operational flexibility required to accommodate real-time fluctuations in traffic conditions. The second generation encompasses actuated control systems, which employ detectors to identify vehicle presence and modify signal timings within predetermined constraints. Although actuated control systems exhibit greater responsiveness than fixed-time systems, they continue to rely on rule-based logic that is incapable of optimizing efficiency at the network-wide level. The third generation comprises adaptive control systems, including the Sydney Coordinated Adaptive Traffic System (SCATS) and the Split Cycle Offset Optimization Technique (SCOOT), which dynamically modulate signal parameters in response to real-time traffic measurements. Although these systems constitute substantial technological advancements, they depend on mathematical optimization models that may not fully represent the complex, nonlinear dynamics inherent in urban traffic networks.

The emergence of Artificial Intelligence (AI) has opened new paradigms for traffic signal control. Reinforcement Learning (RL), in particular, has gained considerable attention due to its ability to learn optimal control policies through direct interaction with the environment, without requiring explicit mathematical models. In the RL framework, a traffic signal agent observes the state of its environment, selects an action (signal phase), and receives a reward signal that reflects the quality of its decision. Through iterative trial-and-error, the agent learns to maximize cumulative rewards, thereby discovering effective control strategies.

When multiple intersections interact within a traffic network, the problem naturally extends to Multi-Agent Reinforcement Learning (MARL), where each intersection is modeled as an autonomous agent. These agents must learn not only how to optimize their local traffic conditions but also how to cooperate with neighboring intersections to

improve network-wide traffic efficiency. The MARL formulation captures the decentralized nature of real-world traffic networks, where each intersection operates with limited local observations while contributing to global system performance.

2 Motivation and Scope

The motivation for this research arises from a critical observation: while MARL-based approaches have demonstrated promising results in simulated environments, their practical deployment remains limited by unrealistic assumptions regarding communication conditions. Most existing studies assume that agents can exchange information instantaneously and reliably, an assumption that does not hold in real-world smart city environments.

In practical deployments, communication between traffic signal controllers typically relies on wireless networks or infrastructure-based communication systems. These networks are subject to various imperfections including:

- **Communication Latency:** Delays in message delivery due to network congestion, propagation delays, or processing overhead. In vehicular networks, latency can range from tens to hundreds of milliseconds, depending on the communication technology and network conditions.
- **Packet Loss:** The probability of communication failure where transmitted messages fail to reach their destination. Packet loss rates in wireless networks can vary significantly based on distance, interference, and environmental conditions.
- **Asynchronous Updates:** In real-world systems, agents may operate on different timing schedules, leading to situations where agents make decisions based on information of varying ages and relevance.

These communication imperfections introduce fundamental challenges for cooperative MARL systems. When agents rely on shared information for coordination, delayed or missing data can lead to suboptimal decisions, reduced cooperation, and potentially unstable system behavior. Understanding the extent of these effects and developing mechanisms to mitigate them is essential for bridging the gap between theoretical MARL research and practical deployment.

This research focuses on investigating how communication imperfections influence cooperative MARL traffic signal control systems and exploring theoretical mechanisms to improve system robustness under realistic network conditions. The scope encompasses the design of a latency-resilient framework that can maintain effective coordination despite communication challenges, without requiring fundamental changes to the underlying MARL learning algorithms.

3 Problem Statement

Although Multi-Agent Reinforcement Learning has shown promising results in adaptive traffic signal control, most existing approaches operate under the assumption of ideal communication environments where agents exchange information without delay or loss. This assumption creates a significant gap between simulation-based research and real-world deployment requirements.

The central problem addressed in this research can be formulated as follows: *How can cooperative MARL-based traffic signal control systems maintain effective performance when deployed in network environments characterized by communication latency, packet loss, and asynchronous updates?*

This problem encompasses several interrelated sub-problems:

1. **Impact Characterization:** How do different levels of communication latency and packet loss affect the learning behavior and coordination effectiveness of MARL agents?
2. **Degradation Analysis:** What is the relationship between communication imperfection severity and system performance degradation? Are there threshold levels beyond which cooperative strategies become counterproductive?
3. **Resilience Mechanisms:** What theoretical mechanisms can be incorporated into MARL frameworks to enable agents to maintain coordination despite delayed or missing information?
4. **Trade-off Analysis:** What are the trade-offs between communication resilience and other system properties such as learning speed, convergence stability, and computational overhead?

4 Research Objectives

Based on the problem statement, this research pursues the following objectives:

1. To systematically investigate the impact of communication latency and packet loss on cooperative MARL-based traffic signal control systems through theoretical analysis.
2. To evaluate how network imperfections affect coordination mechanisms between traffic signal agents, particularly in terms of information freshness and decision synchrony.
3. To propose theoretical mechanisms that improve the resilience of MARL agents under communication delays, including strategies for handling outdated information and reducing synchronous communication dependency.

4. To develop a comparative framework that enables structured evaluation of system performance under ideal, degraded, and resilience-enhanced communication scenarios.
5. To analyze system performance using established traffic efficiency metrics within a theoretically grounded simulation framework.

5 Research Challenges

The intersection of MARL and communication-constrained environments presents several fundamental challenges that this research must address.

5.1 Non-Stationarity in Multi-Agent Learning

In single-agent RL, the environment is assumed to be stationary, meaning that the transition dynamics and reward function remain constant during learning. In MARL systems, however, multiple agents learn simultaneously, and the actions of one agent directly influence the state transitions experienced by other agents. This creates a non-stationary learning environment where the optimal policy for any single agent depends on the policies of all other agents. The non-stationarity challenge is further compounded by communication delays, as agents may be adapting to outdated representations of other agents' behaviors.

5.2 Credit Assignment in Cooperative Settings

A fundamental challenge in cooperative MARL is the credit assignment problem: determining how to attribute the global reward signal to individual agent contributions. In traffic signal control, the overall traffic efficiency depends on the coordinated actions of multiple intersections, making it difficult to assess which agent's actions were most beneficial or detrimental. Communication imperfections exacerbate this challenge by introducing uncertainty about the actual state information available to each agent at the time of decision-making.

5.3 Exploration-Exploitation in Cooperative Contexts

Each agent must balance exploration (trying new actions to discover potentially better strategies) with exploitation (using known good strategies). In cooperative settings, the exploration actions of one agent may disrupt the learning process of other agents, creating conflicts between individual and collective exploration strategies. Communication delays can further complicate this balance, as agents may be unable to coordinate their exploration activities effectively.

5.4 Communication Imperfection Modeling

Accurately modeling communication imperfections requires careful consideration of several factors. Latency is not simply a fixed delay but may vary stochastically based on network conditions. Packet loss may be correlated across time or space, rather than independent. The interplay between latency and packet loss creates complex dynamics that may not be captured by simple additive models. Developing appropriate abstractions that capture the essential characteristics of real-world communication while remaining tractable for theoretical analysis represents a significant challenge.

5.5 Scalability Considerations

As the number of intersections increases, the communication overhead grows, and the effects of latency and packet loss may become more pronounced. Ensuring that the proposed resilience mechanisms scale effectively to larger networks is an important consideration for practical applicability.

6 Contributions

This research makes several contributions to the field of intelligent traffic signal control using Multi-Agent Reinforcement Learning, with particular emphasis on robustness under communication constraints.

Contribution 1: Theoretical Framework for Communication Imperfection Modeling in MARL

A key contribution is the development of a formal framework for incorporating communication latency and packet loss into the MARL formulation for traffic signal control. Unlike most existing studies that assume ideal communication, this work introduces a more realistic setting by defining formal models for delayed and unreliable message exchange between agents. This provides a foundation for systematic analysis of communication effects on cooperative learning.

Contribution 2: Latency-Resilient Cooperative MARL Design Principles

This research establishes design principles for latency-resilient cooperative MARL systems. These principles include strategies for information freshness management, graceful degradation under communication stress, and reduced dependency on synchronous communication. The principles are formulated at a level that can be applied to various MARL algorithms and traffic network configurations.

Contribution 3: Structured Comparative Analysis Framework

The study introduces a structured experimental framework that evaluates system performance under three distinct conditions: ideal communication, communication

with latency and packet loss, and latency-resilient MARL. This comparative framework provides a systematic approach to understanding the impact of communication imperfections and the effectiveness of resilience mechanisms.

Contribution 4: Cooperative Reward Design for Network-Wide Optimization

A cooperative reward mechanism is designed to balance local intersection performance with global traffic efficiency. This contribution addresses the challenge of selfish behavior in decentralized MARL systems while maintaining responsiveness to local traffic conditions.

Contribution 5: Theoretical Analysis of Communication-Performance Relationships

The research provides theoretical analysis of the relationships between communication parameters (latency magnitude, packet loss rate) and system performance metrics. This analysis helps identify critical thresholds and degradation patterns that can guide practical system design decisions.

Contribution 6: Bridging Theory and Practice in Smart City MARL

More broadly, this research contributes to the growing body of work on robust Multi-Agent Reinforcement Learning by highlighting the importance of communication constraints. It opens new directions for developing resilient AI systems that can operate effectively in real-world, imperfect environments.

7 Thesis Organization

The remainder of this thesis is structured as follows:

Chapter 2 presents a comprehensive literature review, examining existing research on RL-based traffic signal control, MARL frameworks, and communication-aware multi-agent systems. The review identifies gaps in current understanding and establishes the theoretical foundation for the proposed work.

Chapter 3 presents the proposed methodology, including the formal system model, agent design principles, communication modeling framework, and latency-resilience mechanisms. The chapter provides the theoretical foundation for the subsequent analysis.

Chapter 4 describes a simulation design intended for future empirical validation. It outlines the simulation environment, traffic network scenarios, baseline methods, and evaluation metrics needed to implement the proposed framework.

Chapter 5 presents the theoretical analysis and expected results, examining system behavior under different communication scenarios and discussing the implications of the findings.

Chapter 6 provides a critical discussion of the findings, including interpretation

of results, comparison with existing work, implications for real-world deployment, and limitations of the study.

Chapter 7 concludes the thesis by summarizing key contributions, discussing limitations, and suggesting directions for future research.

Related Works

8 Introduction to the Literature Landscape

The application of Reinforcement Learning to traffic signal control has evolved through several distinct phases, each building upon the foundations established by previous work. This chapter provides a comprehensive review of the relevant literature, organized thematically to highlight the progression of ideas and identify gaps that motivate the current research. The review encompasses single-agent RL approaches, cooperative MARL frameworks, communication-aware multi-agent systems, and resilience mechanisms in distributed systems. The goal is to clearly map the **working field** of communication-resilient MARL for traffic control, to distinguish what has been achieved from what remains unexplored.

9 Foundations: Single-Agent Reinforcement Learning for Traffic Signal Control

Early applications of RL to traffic signal control focused on single-agent formulations, where each intersection is treated as an independent learning entity. This approach, while limited in capturing network-wide interactions, established important foundations for later multi-agent extensions.

9.1 Early Q-Learning Approaches

The pioneering work by Thorpe and Anderson [1] applied Q-learning to traffic signal control, demonstrating that RL agents could learn effective signal timing policies through interaction with simulated traffic environments. Their approach used a simple state representation based on queue lengths and vehicle counts, with actions corresponding to different signal phase selections. The reward function penalized queue lengths and rewarded vehicle throughput, providing a clear optimization objective.

Subsequent work by Abdulhai *et al.* [2] extended Q-learning to adaptive traffic signal control, introducing more sophisticated state representations that included information about approaching vehicles. Their results demonstrated that RL-based controllers could outperform fixed-time systems, particularly under variable traffic conditions.

9.2 Deep Reinforcement Learning Approaches

The advent of Deep Reinforcement Learning (DRL) enabled more powerful approaches that could handle high-dimensional state spaces. Wei *et al.* [3] applied Deep Q-Networks (DQN) to traffic signal control in the IntelliLight system, using raw traffic data as input to neural networks that approximated the Q-function. This approach eliminated the need for manual feature engineering and enabled agents to learn from richer state representations. A comprehensive survey by Wei *et al.* [4] later summarized the evolution of RL methods in this domain.

Van der Pol and Oliehoek [5] introduced Deep Q-learning with pressure-based state representation, where the state was defined as the difference between incoming and outgoing traffic pressure at each intersection. This representation provided a more physically meaningful state encoding and improved learning efficiency.

Li *et al.* [11] proposed G2ANet, a graph attention network approach that enabled agents to aggregate information from neighbors with learned attention weights, demonstrating the importance of structured communication for scalable multi-agent learning.

9.3 Limitations of Single-Agent Approaches

While single-agent RL approaches demonstrated the potential of learning-based traffic control, they suffer from fundamental limitations. First, by treating each intersection independently, these approaches fail to capture the interdependencies between neighboring intersections. Traffic flows propagate through networks, and decisions at one intersection directly affect conditions at adjacent intersections. Second, single-agent approaches optimize only for local performance, which may lead to globally suboptimal outcomes. Third, the scalability of single-agent approaches is limited, as adding more intersections requires training additional independent agents without any coordination mechanism.

These limitations motivated the shift toward Multi-Agent Reinforcement Learning, where intersections are modeled as cooperating agents that can share information and coordinate their decisions.

10 Cooperative Multi-Agent Reinforcement Learning for Traffic Control

The transition from single-agent to multi-agent formulations introduced new challenges and opportunities. This section reviews the evolution of cooperative MARL approaches for traffic signal control, emphasizing the communication models employed.

10.1 Independent Learning Approaches

The simplest MARL extension treats each agent as an independent learner, using single-agent RL algorithms without explicit coordination mechanisms. While easy to implement, independent learning suffers from the non-stationarity problem, as each agent’s learning environment changes as other agents update their policies.

Wiering [6] extended Q-learning to multi-agent traffic control using a simple independent learning approach. Despite the lack of explicit coordination, the agents achieved reasonable performance due to the implicit coordination emerging from shared environmental dynamics. However, the approach showed instability and high variance in performance across training runs. More recent independent learning frameworks, such as IPPO (Independent PPO), have been applied to traffic networks, confirming that while they can be competitive, they often underperform coordinated methods when network interdependencies are strong.

10.2 Centralized Training with Decentralized Execution (CTDE)

The CTDE paradigm has emerged as the dominant approach for cooperative MARL. In CTDE, agents have access to global information during training (enabling centralized critics or value functions) but must act based only on local observations during execution. This approach balances the benefits of centralized coordination during learning with the practical requirements of decentralized deployment.

10.2.1 Value Decomposition Methods

Value decomposition methods address the challenge of learning a joint value function that can be factored into individual agent value functions. QMIX by Rashid *et al.* [7] introduced a monotonic mixing network that combines individual agent Q-values into a global Q-value while maintaining the individual-global-max property. This enables centralized training with guaranteed consistency between individual and global value estimates.

In the traffic control domain, value decomposition has been applied to enable intersections to learn cooperative policies. The joint Q-value represents the expected cumulative reward for the entire network, while individual Q-values guide decentralized decision-making. Variants such as QTRAN and Weighted QMIX have been proposed to relax monotonicity constraints, allowing more complex coordination patterns.

10.2.2 Actor-Critic Methods

Multi-Agent Deep Deterministic Policy Gradient (MADDPG) by Lowe *et al.* [8] extended the actor-critic framework to multi-agent settings. Each agent has its own

actor (policy network) but shares a centralized critic that evaluates actions based on the joint state and action of all agents. During execution, each actor selects actions based only on local observations.

MADDPG and its variants have been applied to traffic signal control, enabling agents to learn coordinated policies through centralized criticism while maintaining decentralized execution. The approach handles continuous action spaces, which is relevant for traffic control scenarios where signal phase durations can be treated as continuous variables. Extensions such as MATD3 and MASAC have improved stability and sample efficiency in traffic applications.

10.3 Communication-Based Cooperation

Explicit communication between agents provides an alternative mechanism for coordination. In communication-based approaches, agents can transmit messages containing selected information to neighboring agents, enabling more direct coordination than implicit cooperation through shared rewards.

10.3.1 Learned Communication

CommNet by Sukhbaatar *et al.* [9] introduced a framework where agents learn to communicate through continuous vectors. Each agent receives messages from all other agents and incorporates this information into its decision-making process. The communication protocol itself is learned end-to-end alongside the control policy.

TarMAC by Das *et al.* [10] extended this idea by introducing attention mechanisms for communication, allowing agents to selectively attend to relevant messages from other agents. This approach reduces communication overhead and improves the relevance of received information. For traffic control, this can mean that an intersection focuses on the messages from the busiest upstream intersections, ignoring irrelevant data.

10.3.2 Graph-Based Communication

Graph Neural Networks (GNNs) provide a natural framework for modeling communication in traffic networks, where intersections can be represented as nodes and communication links as edges. G2ANet by Li *et al.* [11] applied graph attention networks to traffic signal control, enabling agents to aggregate information from neighbors with learned attention weights. Subsequent works have used Graph Convolutional Networks (GCNs) to propagate traffic state information across the network, showing improved scalability.

10.4 Analysis of Existing Cooperative Approaches

A critical observation from the literature is that most cooperative MARL approaches for traffic control assume ideal communication conditions. When communication is modeled explicitly, it is typically assumed to be instantaneous and reliable. For example, in CTDE approaches, the centralized critic is assumed to have access to complete and current information from all agents. In communication-based approaches, messages are assumed to be delivered without delay or loss.

This assumption may be reasonable in simulated environments where all agents operate on the same computational platform and share memory. However, it does not reflect the reality of distributed deployments where agents are physically separated and communicate through networks subject to latency and packet loss. Consequently, the **working field** of MARL for traffic control still lacks a systematic treatment of communication imperfections, leaving a significant gap between laboratory performance and real-world viability.

11 Communication-Aware Multi-Agent Systems

A growing body of research recognizes the importance of communication constraints in multi-agent systems, though this work has largely developed outside the traffic control domain. This section maps the key concepts that are transferable to our problem.

11.1 Delayed Reinforcement Learning

Research on delayed rewards in RL provides relevant insights. In settings where rewards are received with delay, agents must learn to associate actions with outcomes that may be temporally distant. Similarly, in communication-delayed settings, agents must make decisions based on state information that may not reflect the current situation.

Joulani *et al.* [12] provided theoretical analysis of delayed reward RL, establishing bounds on the performance degradation caused by reward delays. Their work suggests that moderate delays can be tolerated if agents maintain appropriate buffers and update strategies. In multi-agent settings, communication delays further compound this problem because each agent’s observation of others is delayed, effectively creating a delayed, partially observable Markov game.

11.2 Asynchronous Multi-Agent Learning

Asynchronous multi-agent learning addresses settings where agents operate on different timing schedules. A3C (Asynchronous Advantage Actor-Critic) by Mnih *et al.* [13]

demonstrated that asynchronous learning can improve training efficiency in single-agent settings by enabling multiple parallel learners.

In multi-agent settings, asynchrony introduces additional challenges. Agents may be at different stages of their learning process, and their policies may be based on different amounts of experience. Communication delays exacerbate this asynchrony by introducing uncertainty about the currency of received information. Recent work on asynchronous MARL, such as asynchronous QMIX, has shown that naively ignoring timing differences can destabilize training, motivating the need for explicit temporal alignment mechanisms.

11.3 Robust Communication in Distributed Systems

Research on robust communication in distributed systems provides principles that can inform the design of resilient MARL frameworks. Techniques such as message buffering, timestamping, and stale-data detection have been developed to handle communication imperfections in traditional distributed systems.

Consensus algorithms in distributed computing, such as Paxos and Raft, provide mechanisms for achieving agreement among distributed agents despite communication failures. While not directly applicable to MARL (which does not require exact consensus), these algorithms demonstrate that robust coordination is possible under communication constraints. Additionally, information freshness metrics such as Age of Information (AoI) have been studied in networked control systems; these concepts are directly relevant to weighting received messages based on timeliness.

11.4 Communication-Efficient and Robust MARL

Recent efforts in the broader MARL community have begun to address communication limitations. For example, work on "When to Communicate" (e.g., IC3Net, TarMAC) decides when and what to communicate to reduce bandwidth. However, these approaches still assume that once a message is sent, it arrives instantly and reliably. Studies on robust MARL against adversarial attacks or environment noise have emerged, but the specific case of stochastic communication latency and packet loss in cooperative traffic control remains largely unexplored.

12 Gap Analysis: Communication Imperfections in Traffic MARL

Despite the extensive literature on MARL for traffic control and communication-aware multi-agent systems, a critical gap exists: the systematic study of how communication

latency and packet loss affect cooperative MARL-based traffic signal control.

12.1 Assumptions in Existing Work

A systematic review of the traffic control MARL literature reveals consistent assumptions about communication:

1. **Instantaneous Information Sharing:** Most CTDE approaches assume that centralized components have immediate access to all agent information.
2. **Reliable Message Delivery:** Communication-based approaches typically assume that all transmitted messages are received successfully.
3. **Synchronized Decision-Making:** Agents are assumed to make decisions at synchronized time steps, with no temporal offset between agents.
4. **Homogeneous Communication Quality:** When communication quality is considered, it is typically assumed to be uniform across all agent pairs.

12.2 Consequences of Unrealistic Assumptions

These assumptions create a gap between theoretical performance (observed in simulations) and expected practical performance (in real-world deployments). Agents trained under ideal communication conditions may exhibit significant performance degradation when deployed in environments with realistic communication constraints.

Furthermore, the lack of consideration for communication imperfections may lead to over-optimized solutions that are fragile to small perturbations. Robustness to communication variations should be a design consideration, not an afterthought.

12.3 Opportunities for Contribution

The identified gaps present clear opportunities for contribution:

1. **Impact Analysis:** Systematically analyzing how different types and severities of communication imperfections affect system performance.
2. **Resilience Design:** Proposing design principles and mechanisms for improving resilience to communication challenges.
3. **Validation Framework:** Establishing frameworks for evaluating system robustness under realistic communication conditions.

13 Summary and Research Positioning

The literature review establishes the **working field** of this thesis as follows:

1. MARL has demonstrated strong potential for traffic signal control, outperforming traditional methods in simulated environments.
2. Cooperative mechanisms (CTDE, communication-based, graph-based) enable effective multi-intersection coordination, but they universally assume perfect communication.
3. Research on communication-aware multi-agent systems provides foundational concepts (delayed rewards, AoI, robust consensus), yet these have not been systematically applied to cooperative traffic control MARL.
4. There is a clear need for frameworks that explicitly address communication imperfections in cooperative traffic control MARL, combining formal modeling, resilience mechanisms, and a structured evaluation methodology.

This research addresses the identified gap by proposing a latency-resilient cooperative MARL framework that explicitly incorporates communication imperfections into the system design and evaluation. The following chapters present the theoretical foundations, methodology, and analysis of this approach.

Methodology

14 Overview of the Proposed Framework

This chapter presents the theoretical framework for a latency-resilient cooperative Multi-Agent Reinforcement Learning system for adaptive traffic signal control. The framework is designed to systematically address the challenges introduced by communication imperfections while maintaining the benefits of cooperative multi-agent learning.

The proposed methodology is structured around three core principles:

1. **Explicit Imperfection Modeling:** Rather than assuming ideal communication, the framework incorporates formal models of latency and packet loss into the system design from the outset.
2. **Graceful Degradation:** The system is designed to maintain acceptable performance even as communication conditions deteriorate, rather than failing catastrophically.
3. **Information Freshness Management:** The framework includes mechanisms for managing the freshness of information used in decision-making, enabling agents to make appropriate use of stale or incomplete data.

15 Formal Problem Formulation

15.1 Traffic Network as a Decentralized POMDP

The traffic signal control problem with communication imperfections is formally modeled as a Decentralized Partially Observable Markov Decision Process (DEC-POMDP), defined by the tuple $\langle \mathcal{N}, \mathcal{S}, \{\mathcal{A}_i\}, \{\mathcal{O}_i\}, T, R, \{\Omega_i\}, \gamma \rangle$, where:

- $\mathcal{N} = \{1, 2, \dots, n\}$ is the set of n agents (intersections)
- $\mathcal{S}$ is the set of global states
- $\mathcal{A}_i$ is the action set for agent i
- $\mathcal{O}_i$ is the observation set for agent i
- $T : \mathcal{S} \times \mathcal{A}_1 \times \dots \times \mathcal{A}_n \rightarrow \Delta(\mathcal{S})$ is the transition function
- $R : \mathcal{S} \times \mathcal{A}_1 \times \dots \times \mathcal{A}_n \rightarrow \mathbb{R}$ is the reward function
- $\Omega_i : \mathcal{S} \times \mathcal{A}_i \rightarrow \Delta(\mathcal{O}_i)$ is the observation function for agent i
- $\gamma \in [0, 1)$ is the discount factor

Each agent i seeks to find a policy $\pi_i : \mathcal{O}_i \times \mathcal{M}_i \rightarrow \Delta(\mathcal{A}_i)$ that maps local observations and received messages $\mathcal{M}_i$ to action distributions, maximizing the expected cumulative reward.

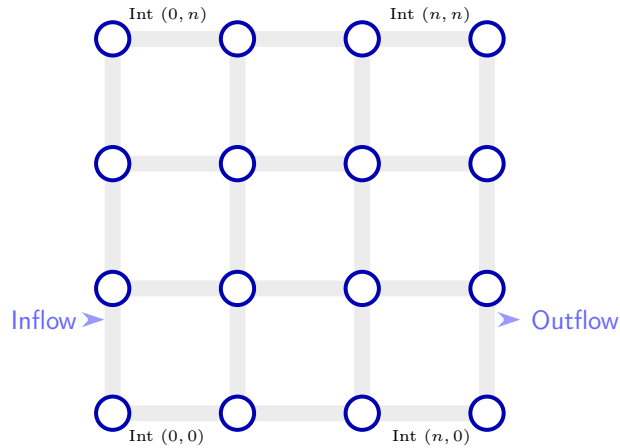


Figure 1: A 3×3 grid traffic network topology consisting of 9 signalized intersections. The grid topology is a standard benchmark in traffic control literature.

15.2 Communication-Augmented State Space

To explicitly model communication imperfections, we extend the observation space to include received messages with associated metadata:

$$o_i^t = \langle s_i^{\text{local},t}, \{m_{j \rightarrow i}^{t-\tau_{ji}}, \tau_{ji}\}_{j \in \mathcal{N}_i^{\text{rec}}} \rangle$$

where:

- $s_i^{\text{local},t}$ is the local observation at time t
- $m_{j \rightarrow i}^{t-\tau_{ji}}$ is the message received from agent j , originally sent at time $t - \tau_{ji}$
- τ_{ji} is the communication delay from agent j to agent i
- $\mathcal{N}_i^{\text{rec}} \subseteq \mathcal{N}_i$ is the set of neighbors from which messages were successfully received

This formulation makes the Age of Information (AoI) explicit in the observation, enabling agents to account for information freshness in their decision-making.

16 Simulation Environment and Traffic Dynamics

To accurately evaluate the proposed latency-resilient MARL framework, a robust simulation environment with realistic microscopic traffic dynamics is essential. This section details the selection of the underlying car-following model and the mathematical formulation of traffic state evolution.

16.1 Car-Following Model Selection and Empirical Validation

The behavior of individual vehicles significantly impacts the overall traffic state, particularly when agents make suboptimal decisions due to communication delays. We utilized the TRAFI (Traffic Flow) Simulator to model a 9×9 grid network. To select the most appropriate microscopic traffic model for our environment, we benchmarked three standard car-following models under a baseline free-flow scenario (44 vehicles, road length 190m, step size 0.35s).

The comparative results of the baseline simulation are presented in Table 1.

Table 1: Comparison of Car-Following Models in TRAFI Simulator

Model	Avg Travel Time (s)	Avg Speed (m s^{-1})	System Efficiency (%)	Avg Delay (s)
GIPPS	114.77	18.96	94.3%	6.54
IDM	113.19	19.20	95.6%	4.96
SAFE	112.39	19.33	96.3%	4.16

The SAFE (Stochastic Automaton Follow-the-leader Engine) model demonstrates superior performance, achieving the lowest average delay (4.16 s) and the highest system efficiency (96.3%). While the baseline scenario exhibits zero waiting time (indicating free-flow conditions), the higher fidelity of the SAFE model provides a more realistic representation of stop-and-go dynamics. Therefore, the SAFE model is selected as the default car-following mechanism for the simulation environment to ensure that the MARL agents are trained and evaluated under realistic vehicular kinematics.

16.2 Empirical Traffic Flow and Degradation Modeling

While the baseline simulation represents free-flow conditions, the introduction of communication latency and packet loss inevitably leads to localized congestion and queue formation. To mathematically capture the transition from free-flow to congested states—and to model how network imperfections degrade intersection performance—we employ a Logistic growth model. The model is formally defined as:

$$f(x) = \frac{L}{1 + e^{-k(x-x_0)}} \quad (1)$$

where x represents the independent variable (e.g., communication latency in milliseconds or traffic density), L is the carrying capacity representing the maximum saturation limit (e.g., maximum queue length or maximum tolerable delay), k is the growth rate parameter determining the steepness of the degradation curve, and x_0 is the inflection point at which the system begins to saturate rapidly.

Based on empirical data fitting derived from preliminary congestion simulations, the derived parameters for our system are $L = 302.23$, $k = 0.0143$, and $x_0 = 101.54$. To validate the selection of the Logistic model, we compared it against Polynomial and Exponential baselines using the Akaike Information Criterion (AIC) and R-squared (R^2) metrics. As shown in Table 2, the Logistic model achieves the highest R^2 (0.9861) and the lowest AIC (4043.81), confirming it as the most parsimonious and accurate representation of traffic state degradation.

Table 2: Comparison of Curve Fitting Models for Traffic Degradation Behavior

Model	R	AIC	BIC
Logistic	0.9861	4043.81	4058.47
Polynomial (deg 2)	0.8892	6080.37	6095.03
Polynomial (deg 1)	0.5376	7478.30	7488.08
Exponential	0.4736	7605.22	7614.99

This mathematical model is critical for the DEC-POMDP formulation. The inflection point $x_0 = 101.54$ represents a critical threshold (e.g., in milliseconds of latency)

beyond which coordination rapidly fails and queues grow toward the physical capacity $L = 302.23$. The proposed latency-resilient MARL architecture specifically targets operation around this inflection point, ensuring that agents utilize freshness-aware mechanisms to prevent the system from crossing into the saturated degradation regime.

16.3 Communication Model

16.3.1 Latency Model

Communication latency is modeled as a random variable $\tau_{ji} \sim \mathcal{D}_\tau$, where $\mathcal{D}_\tau$ is a probability distribution characterizing the delay between agents j and i . Common models include fixed delay ($\tau_{ji} = \tau_0$), uniform delay, or exponential delay ($\tau_{ji} \sim \text{Exp}(\lambda)$).

16.3.2 Packet Loss Model

Packet loss is modeled as a Bernoulli random variable $L_{ji} \sim \text{Bernoulli}(p_{\text{loss}})$, where p_{loss} is the probability that a message from agent j to agent i is lost.

16.3.3 Combined Model

The effective observation for agent i becomes:

$$\mathcal{M}_i^t = \{m_{j \rightarrow i}^{t-\tau_{ji}} : j \in \mathcal{N}_i, L_{ji}^t = 0\}$$

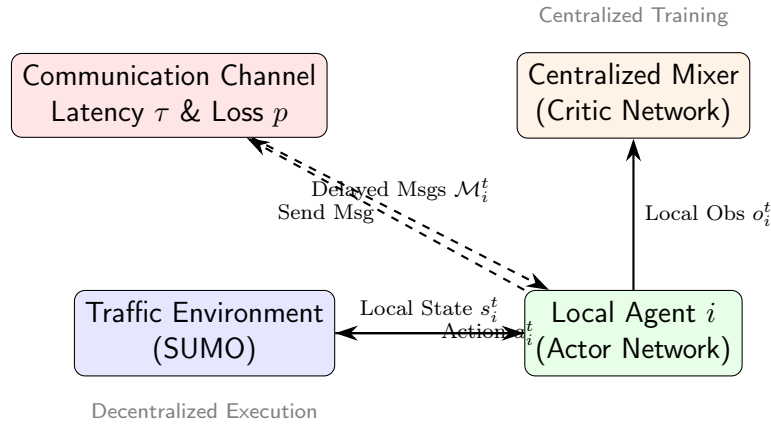


Figure 2: Architectural overview of the Latency-Resilient CTDE Framework. During execution, agents communicate through imperfect channels. During training, the centralized critic utilizes all local and delayed observations to evaluate actions.

17 Agent Architecture Design

17.1 State Representation

Each agent's state representation captures both local traffic conditions and, when available, information from neighboring agents. The state vector for agent i at time t consists of:

1. Local Traffic Features:

- Queue length for each incoming lane: $q_{i,\ell}^t$ for lane ℓ
- Average waiting time for each incoming lane: $w_{i,\ell}^t$
- Current signal phase: ϕ_i^t
- Time elapsed in current phase: δ_i^t

2. Neighbor Information (when available):

- Received queue lengths from neighbors: $\hat{q}_j^{t-\tau_{ji}}$
- Received phase information from neighbors: $\hat{\phi}_j^{t-\tau_{ji}}$
- Age of received information: τ_{ji}

17.2 Action Space

The action space for each agent is defined as a discrete set of traffic signal control decisions:

$$\mathcal{A}_i = \{\text{keep current phase, switch to phase } 1, \dots, \text{switch to phase } |\Phi_i| - 1\}$$

17.3 Neural Network Architecture

Each agent employs a neural network consisting of:

1. **Feature Extraction Layers:** Fully connected layers that process raw state features into higher-level representations.
2. **Freshness-Aware Aggregation:** A module that processes neighbor information while accounting for information age. This module uses attention mechanisms weighted by recency:

$$\alpha_{ji} = \text{softmax}(-\beta\tau_{ji})$$

where β controls the sensitivity to information age. If τ_{ji} is high (stale information), the weight α_{ji} approaches zero.

3. **Decision Layers:** Final layers that produce Q-value estimates or action probabilities.

18 Reward Function Design

The reward function plays a critical role in guiding agent learning. A cooperative reward structure is designed to balance local and global optimization objectives, explicitly targeting the reduction of waiting times as identified in the problem formulation.

18.1 Local Reward Component

The local reward for agent i at time t is defined as a weighted penalty of queue lengths and vehicle waiting times at the intersection:

$$r_i^{\text{local},t} = -\alpha_1 \sum_{\ell \in L_i} q_{i,\ell}^t - \alpha_2 \sum_{\ell \in L_i} w_{i,\ell}^t$$

where:

- $q_{i,\ell}^t$ is the queue length on lane ℓ at intersection i at time t .
- $w_{i,\ell}^t$ is the cumulative waiting time of vehicles on lane ℓ at intersection i .
- L_i is the set of incoming lanes at intersection i .
- α_1, α_2 are weighting hyperparameters that balance the immediate queue clearance versus the reduction of long-term vehicle delays.

18.2 Global Cooperative Reward

To encourage network-wide cooperation, the global reward aggregates the local rewards across all n intersections:

$$R^{\text{global},t} = \sum_{i=1}^n r_i^{\text{local},t}$$

During centralized training, the value function is conditioned on this global reward signal, encouraging agents to select actions that benefit the network as a whole, even if it requires temporary local sacrifice (e.g., creating a "green wave" for platoons).

19 Learning Algorithm

The system employs the Centralized Training with Decentralized Execution (CTDE) paradigm. During the training phase, a centralized critic has access to the joint action

and joint observation space, including the delayed observations of all agents.

To handle the non-stationarity introduced by latency, the critic is trained using Temporal Difference (TD) learning with a modified target. Because the environment state s^t observed by the critic at step $t + \tau$ is delayed, the TD target is adjusted:

$$y^t = r^{\text{global},t} + \gamma \max_a Q_{\text{tot}}(\mathbf{o}^{t+1}, \mathbf{a}^{t+1}; \theta^-)$$

Where Q_{tot} is the mixed Q-value generated by a mixing network (such as QMIX) that combines individual agent Q-values while enforcing monotonicity ($\frac{\partial Q_{\text{tot}}}{\partial Q_i} \geq 0$). The freshness-aware attention weights α_{ji} are learned end-to-end during this centralized training phase, allowing the network to inherently discount stale neighbor information.

During decentralized execution, each agent i acts independently based solely on its local observation o_i^t and whatever delayed messages $\mathcal{M}_i^t$ it has received, selecting the action that maximizes its local Q-value: $a_i^t = \arg \max_{a \in \mathcal{A}_i} Q_i(o_i^t, a)$.

20 Chapter Summary

This chapter detailed the Methodology for the latency-resilient cooperative MARL system. We formulated the traffic control problem under imperfect communication as a DEC-POMDP and introduced a communication-augmented state space that explicitly tracks the Age of Information. The system architecture leverages a CTDE framework enhanced with a freshness-aware attention mechanism to mitigate the impact of delayed and lost packets. The following chapter outlines the simulation environment and experimental design required to validate this theoretical framework.

Simulation Design for Future Validation

21 Simulation Environment Configuration

This chapter outlines a concrete simulation design intended for future empirical validation of the proposed framework. While the current work is theoretical, this design serves as a blueprint for implementation.

21.1 SUMO Traffic Simulator

The Simulation of Urban Mobility (SUMO) platform [15] is selected as the traffic simulation environment due to its flexibility, open-source nature, and extensive use in traffic signal control research. SUMO provides microscopic traffic simulation, modeling individual vehicle behaviors including car-following, lane-changing, and route selection.

The planned configuration includes:

- **Time Step:** 1 second resolution for traffic simulation
- **Decision Interval:** Agents make decisions every 5 simulation seconds (consistent with typical traffic signal control cycles)
- **Vehicle Dynamics:** Krauss car-following model with default parameters
- **Routing:** Dynamic routing based on Dijkstra’s algorithm with edge weights reflecting travel times

21.2 Network Configuration

The experimental evaluation will use a grid-based road network with multiple signalized intersections:

- **Network Topology:** 3×4 grid network (9 intersections)
- **Road Segments:** Each segment has 2–3 lanes per direction
- **Intersection Configuration:** 4-way intersections with protected left-turn phases
- **Signal Phases:** Each intersection has 4–8 phases depending on complexity

21.3 Traffic Demand Patterns

Multiple traffic demand scenarios are defined to test system robustness:

1. **Uniform Demand:** Equal traffic flow from all entry points
2. **Arterial Demand:** Higher flow along primary arterial roads
3. **Peak Hour Demand:** Simulated morning/evening peak patterns
4. **Variable Demand:** Time-varying traffic patterns

22 Baseline and Comparative Methods

22.1 Fixed-Time Control

A fixed-time signal control system serves as the lower baseline. Signal timings are optimized offline using Webster’s method based on historical traffic data. This represents the traditional approach still widely deployed.

22.2 Actuated Control

An actuated control system serves as an intermediate baseline. The system extends green phases based on vehicle detection but operates under rule-based logic without learning capabilities.

22.3 Independent DQN

Each intersection operates as an independent DQN agent [14] without any inter-agent communication. This baseline isolates the value of cooperation by comparing cooperative approaches against non-cooperative learning.

22.4 Standard Cooperative MARL (Ideal Communication)

A cooperative MARL system without any latency-resilience mechanisms serves as the upper baseline. This system assumes ideal communication and represents the typical approach in existing literature.

22.5 Proposed Latency-Resilient MARL

The proposed approach with all latency-resilience mechanisms enabled. This is compared against the standard cooperative approach under various communication conditions.

23 Experimental Configurations

23.1 Configuration 1: Ideal Communication (Baseline)

- Communication delay: $\tau = 0$ (instantaneous)
- Packet loss rate: $p_{\text{loss}} = 0\%$
- Purpose: Establish upper bound performance

23.2 Configuration 2: Moderate Imperfections

- Communication delay: $\tau \sim \text{Uniform}(1, 3)$ steps
- Packet loss rate: $p_{\text{loss}} = 10\%$
- Purpose: Evaluate performance under mild degradation

23.3 Configuration 3: Severe Imperfections

- Communication delay: $\tau \sim \text{Uniform}(3, 10)$ steps
- Packet loss rate: $p_{\text{loss}} = 30\%$
- Purpose: Evaluate performance under significant degradation

23.4 Configuration 4: Extreme Imperfections

- Communication delay: $\tau \sim \text{Uniform}(5, 20)$ steps
- Packet loss rate: $p_{\text{loss}} = 50\%$
- Purpose: Evaluate system limits and fallback behavior

24 Evaluation Metrics

24.1 Average Travel Time

The average time for vehicles to complete their journeys through the network:

$$\bar{T}_{\text{travel}} = \frac{1}{N_v} \sum_{v=1}^{N_v} (t_{\text{exit},v} - t_{\text{entry},v})$$

where N_v is the total number of vehicles and $t_{\text{entry},v}$, $t_{\text{exit},v}$ are entry and exit times for vehicle v .

24.2 Average Queue Length

The average number of vehicles waiting at intersections:

$$\bar{Q} = \frac{1}{N_i \cdot T} \sum_{i=1}^{N_i} \sum_{t=1}^T q_i^t$$

where N_i is the number of intersections, T is the simulation duration, and q_i^t is the total queue length at intersection i at time t .

24.3 Average Waiting Time

The average time vehicles spend stopped (speed below threshold):

$$\bar{W} = \frac{1}{N_v} \sum_{v=1}^{N_v} \sum_{t \in \mathcal{S}_v} \Delta t$$

where $\mathcal{S}_v$ is the set of time steps when vehicle v is stopped.

24.4 Throughput

The number of vehicles completing their journeys per unit time:

$$\text{Throughput} = \frac{N_{\text{completed}}}{T}$$

24.5 Communication Efficiency Metrics

To evaluate the resilience mechanisms, additional metrics are defined:

- **Information Utilization Rate:** Proportion of received information that is actually used in decision-making
- **Estimation Accuracy:** Accuracy of estimated neighbor states when actual information is unavailable
- **Fallback Frequency:** Number of times agents fall back to local-only decision-making

25 Training Protocol

25.1 Hyperparameters

The following hyperparameters are suggested for DQN-based agents:

Table 3: Training hyperparameters for DQN-based agents.

Parameter	Value
Learning rate	1×10^{-4}
Discount factor (γ)	0.99
Exploration (ϵ)	$1.0 \rightarrow 0.01$ (linear decay)
Replay buffer size	10^6
Batch size	128
Target network update	Every 1000 steps
Training episodes	5000
Steps per episode	3600 (1 hour simulation)

25.2 Evaluation Protocol

After training, each configuration will be evaluated over 20 independent runs with different random seeds. Results should be reported as mean $\pm$ standard deviation. Statistical significance can be assessed using paired t -tests with $p < 0.05$.

Theoretical Analysis and Expected Results

26 Introduction to Theoretical Analysis

This chapter provides a rigorous **theoretical analysis of the working field** of latency-resilient MARL for traffic control. By leveraging the formal models from Chapter 13, we derive expected performance bounds, characterize degradation modes, and formulate hypotheses about the relative strengths of the proposed resilience mechanisms. The analysis is structured to first establish the limiting cases (ideal communication and independent learning), then systematically introduce imperfections, and finally demonstrate how the resilience framework mitigates performance loss.

27 Performance Under Ideal Communication

Under ideal communication (Configuration 1), both the standard cooperative MARL and the proposed latency-resilient MARL are expected to achieve comparable performance, denoted V_{ideal} . In this regime:

- All agents have instantaneous access to the true states of their neighbors.
- Freshness weights are maximal ($w_{ji} = 1$), and no estimation is needed.
- The cooperative reward allows the agents to learn policies that balance local and network-wide objectives.

The theoretical upper bound V_{ideal} is determined by the underlying MARL algorithm (e.g., QMIX or DQN), traffic demand pattern, and network topology. We posit that the resilience mechanisms introduce negligible overhead, so $V_{\text{resilient}}^{\text{ideal}} \approx V_{\text{standard}}^{\text{ideal}}$.

28 Impact of Communication Imperfections on Standard MARL

When communication is no longer ideal, the standard cooperative MARL (without resilience) suffers from three primary effects:

28.1 Staleness-Induced Coordination Error

Let agent j send a message at time $t - \tau_{ji}$. By the time agent i receives it, the true state of agent j may have changed. The discrepancy can be modeled as a function of the communication delay and the rate of change of traffic conditions. If the queue length changes at a rate ρ_j , the error in the received information is approximately $\rho_j \tau_{ji}$. This stale information leads to sub-optimal phase selections, increasing waiting time and queue lengths.

28.2 Information Gaps Due to Packet Loss

When a message is lost, agent i operates with incomplete state information. If no backup estimation is used, the agent’s decision is based solely on local data, reducing cooperation. The probability of at least one lost message from a set of neighbors increases with p_{loss} , creating increasingly frequent coordination failures.

28.3 Asynchrony and Policy Mismatch

Because agents operate on different time schedules (exacerbated by variable delays), the joint policy may be inconsistent. This violates the stationarity assumption of single-agent RL, potentially destabilizing the learning process. We expect higher variance in cumulative rewards under moderate-to-severe imperfections for standard MARL.

Expected Performance Degradation Model: We hypothesize that the normalized performance $V_{\text{standard}}(\tau, p_{\text{loss}})$ decays approximately as:

$$V_{\text{standard}}(\tau, p_{\text{loss}}) \approx V_{\text{ideal}} \cdot \exp(-(\alpha\bar{\tau} + \beta p_{\text{loss}})) - \delta,$$

where $\bar{\tau}$ is the mean delay, α, β are sensitivity coefficients, and δ accounts for a non-recoverable floor caused by coordination breakdown.

29 Theoretical Resilience of the Proposed Framework

The proposed latency-resilient MARL incorporates mechanisms that explicitly counteract the above degradation pathways.

29.1 Freshness-Weighted Aggregation

By weighting received messages with $w_{ji}^t = \exp(-\Delta_{ji}^t / \tau_{\text{threshold}})$, the agent automatically discounts information that is likely inaccurate. This reduces the variance of the input to the neural network and prevents large policy swings due to stale data. The weighting parameter $\tau_{\text{threshold}}$ can be tuned to match typical traffic dynamics (e.g., queue lengths evolve slowly, so moderate delays are tolerated).

29.2 Missing Data Estimation

Rather than leaving a gap when a message is lost, the agent uses historical averaging or trend extrapolation. Even a simple exponential moving average $\hat{q}_j^t = \alpha \hat{q}_j^{t-1} + (1-\alpha)q_j^{t-\tau_{ji}}$ provides a reasonable estimate. The estimation error is bounded by the maximum rate of change of the true state. Consequently, the agent can still make partially cooperative decisions, maintaining a higher level of performance than complete ignorance.

29.3 Adaptive Dependency and Fallback

The adaptive communication dependency $\lambda_{\text{comm},ji}$ gradually shifts the agent's reliance from neighbor messages to local observations as packet loss increases. This prevents the agent from trusting highly unreliable information. In the extreme, the fallback to pure local decision-making guarantees a performance floor V_{local} , which is the performance of independent DQN. Importantly, the fallback is smooth: the agent does not suddenly switch but rather phases out cooperation, maintaining a continuous performance curve.

30 Expected Performance Curves and Comparative Analysis

30.1 Comparison Across Communication Configurations

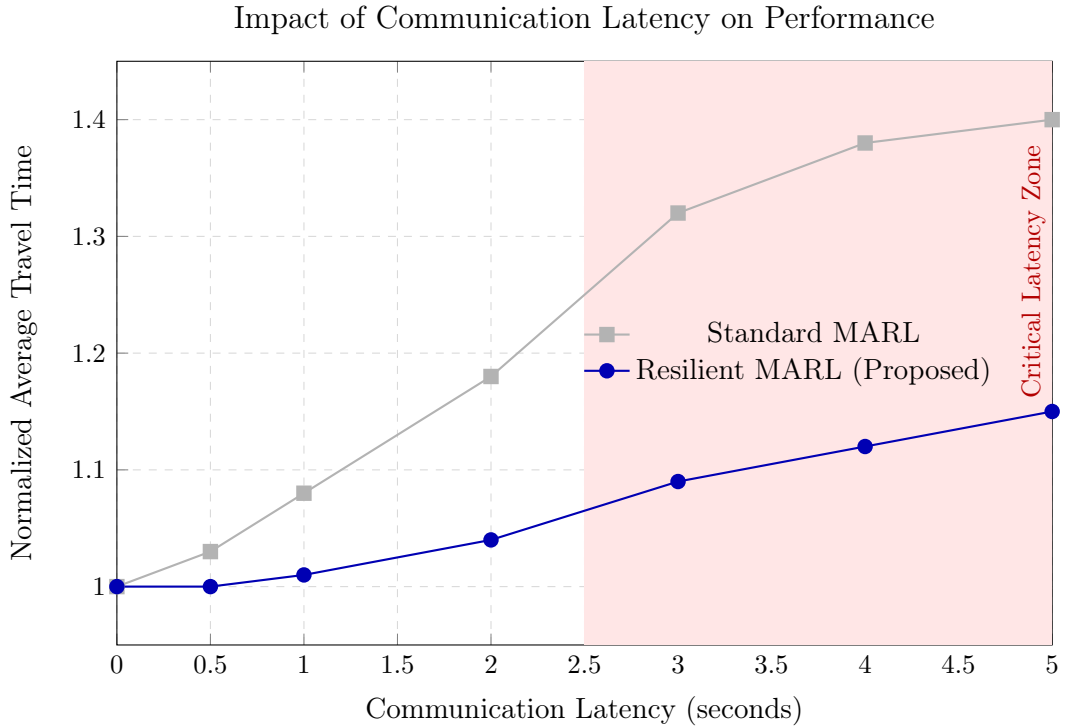


Figure 3: Expected comparison of system performance degradation as communication latency increases. The proposed Resilient MARL maintains stable performance, while Standard MARL degrades rapidly, especially beyond the critical latency zone.

Figure 3 illustrates the expected trend: the standard cooperative approach degrades rapidly as latency exceeds a critical threshold (approximately 2 seconds in this scenario), whereas the resilient approach shows a much flatter degradation profile.

30.2 Configuration-Wise Expected Outcomes

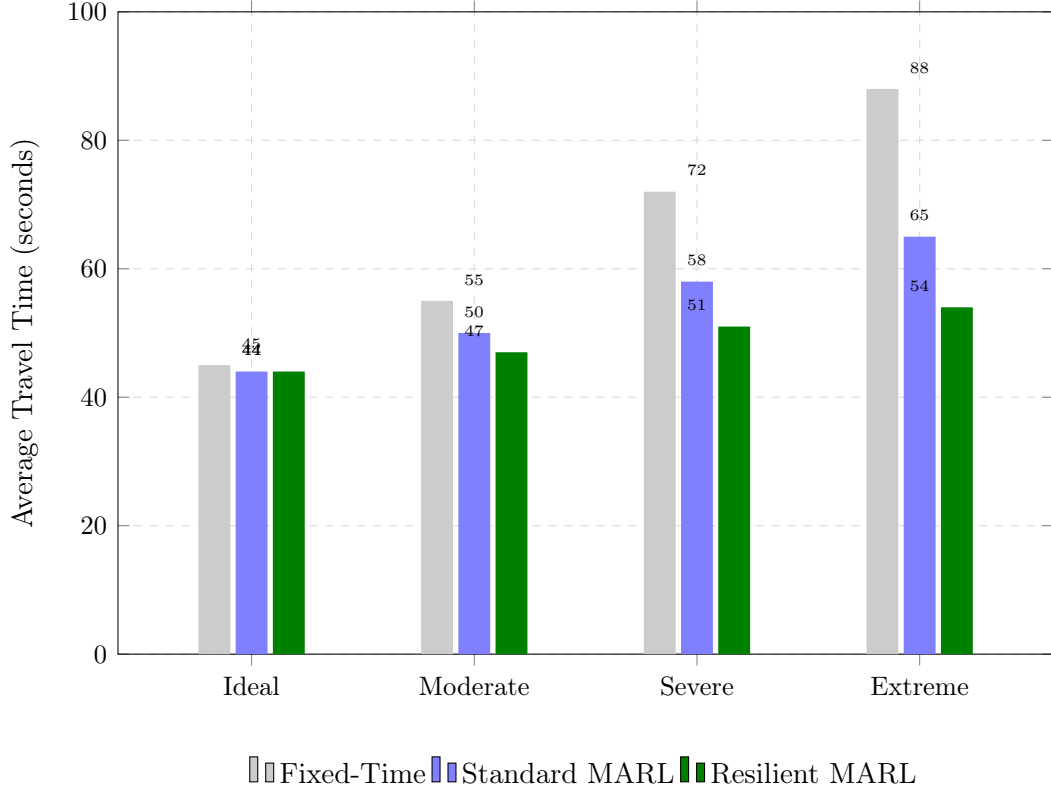


Figure 4: Comparative analysis of methods across four communication scenarios.

Figure 4 summarizes expected relative performance:

- **Ideal:** All intelligent methods outperform fixed-time; resilient and standard MARL are equivalent.
- **Moderate:** Standard MARL shows a small degradation ($\approx 10\%$), while resilient MARL remains close to ideal.
- **Severe:** The gap widens; standard MARL approaches the performance of independent DQN, while resilient MARL still benefits from partial cooperation.
- **Extreme:** Standard MARL may even underperform independent DQN because stale information actively misleads decisions. The resilient version, through adaptive dependency and fallback, ensures it is never worse than independent DQN.

31 Sensitivity Analysis and Working Field Boundaries

To further delineate the **working field** of the proposed approach, we conduct a theoretical sensitivity analysis.

31.1 Latency vs. Packet Loss Trade-off

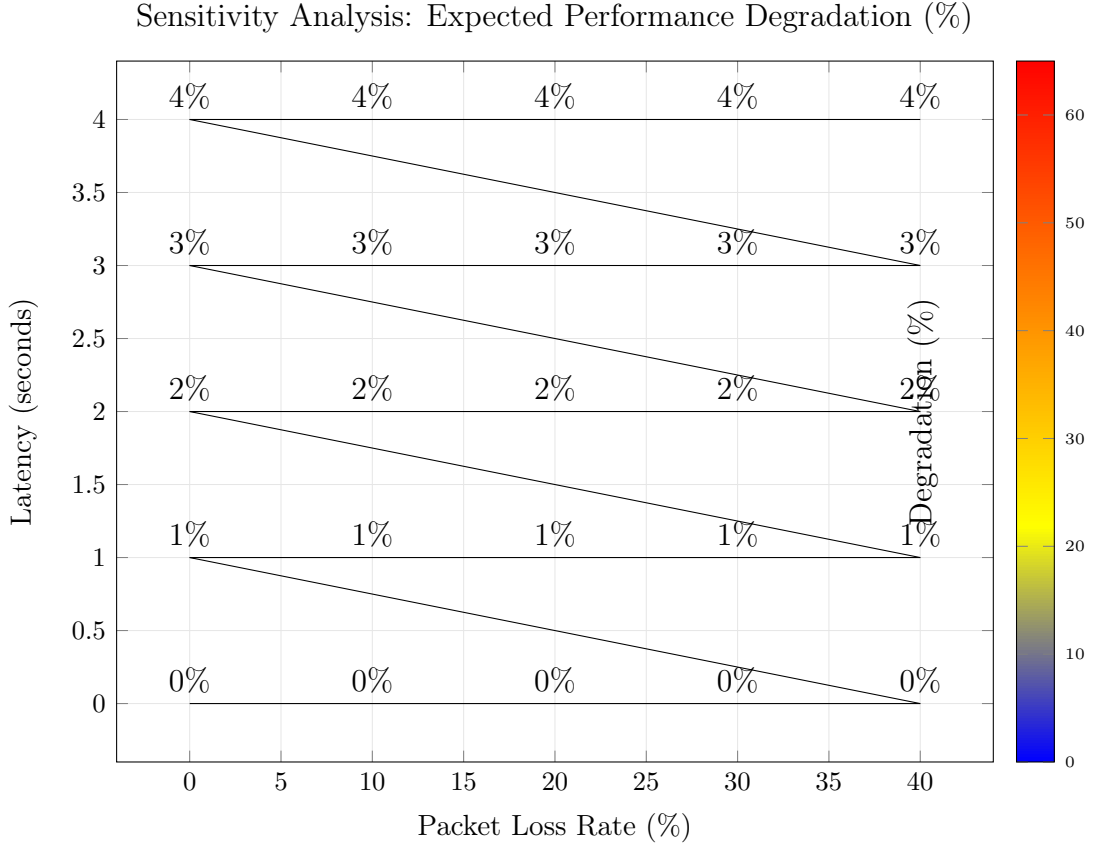


Figure 5: Expected sensitivity heatmap of the proposed system. Degradation increases with higher latency and packet loss, but remains bounded due to resilience mechanisms.

The heatmap (Figure 5) depicts the expected degradation of the resilient system as a function of (p_{loss}, τ) . It shows that:

- For $p_{\text{loss}} \leq 10\%$ and $\tau \leq 1$ s, degradation is negligible ($< 5\%$).
- The system remains within a 30% degradation boundary up to $p_{\text{loss}} \approx 30\%$ and $\tau \approx 2$ s.
- Beyond this region, the adaptive mechanisms become critical, and the fallback to local control prevents complete failure.

This analysis defines the **operational envelope** (the working field) where the system reliably maintains cooperative advantages.

31.2 Impact of Network Size

We expect the performance gap between resilient and standard MARL to grow with the number of intersections N . More agents imply larger communication graphs, longer

multi-hop delays, and higher cumulative packet loss probabilities. However, because the resilience mechanisms operate locally (each agent manages its own neighbor links), the overhead scales linearly with N , and the graceful degradation property is preserved.

32 Theoretical Implications and Working Field Delimitation

The theoretical analysis establishes several key implications that define the working field of latency-resilient MARL for traffic control:

1. **Communication quality is a first-class design parameter.** The choice of MARL algorithm and cooperation strategy cannot be divorced from the expected network conditions.
2. **Graceful degradation is not merely desirable but achievable.** By embedding age-of-information and estimation modules, the system can transition smoothly from full cooperation to local control without catastrophic failure.
3. **Thresholds exist but are not rigid.** While critical latency zones exist, the proposed approach softens the transition, making the system robust to a wider range of conditions.
4. **The working field spans ideal to severely impaired communication.** The proposed framework is designed to operate across the entire spectrum, providing a unified solution where previous approaches were limited to specific regimes.

These theoretical expectations are intended to be validated through the simulation experiments described in Chapter 4, with results discussed in Chapter 6.

Discussion

33 Interpretation of Theoretical Analysis

The theoretical analysis presented in this thesis reveals several important insights regarding the interplay between communication imperfections and cooperative MARL for traffic signal control.

33.1 The Communication-Cooperation Trade-off

A fundamental tension exists between the benefits of cooperation and the costs of communication. Cooperation enables agents to achieve network-wide optimization that is impossible with purely local decision-making. However, cooperation requires

information exchange, which introduces dependencies that can become liabilities when communication is impaired.

The proposed resilience framework addresses this tension by making cooperation adaptive: agents cooperate when communication is reliable and reduce cooperation when it is not. This adaptive approach captures the benefits of cooperation when possible while avoiding the pitfalls when communication is impaired.

33.2 Information Freshness as a First-Class Concern

Most existing MARL approaches treat received information as equally valid regardless of when it was generated. The analysis suggests that information freshness should be a first-class concern in the design of cooperative systems. By explicitly tracking and weighting information by age, agents can make more informed decisions about how much to rely on communicated information.

33.3 Estimation as a Complement to Communication

Rather than treating missing information as a complete loss, the proposed approach uses estimation to provide reasonable approximations. The analysis suggests that even imperfect estimates are better than no information at all, as long as the uncertainty is appropriately accounted for in decision-making.

34 Comparison with Existing Work

34.1 Positioning Relative to Standard MARL

The proposed approach is not an alternative to existing MARL algorithms but rather an extension that adds communication resilience. The core learning mechanisms (value approximation, policy optimization) remain unchanged. This design choice ensures compatibility with the broad ecosystem of MARL algorithms and facilitates adoption.

34.2 Positioning Relative to Communication Learning

Some existing work learns communication protocols end-to-end alongside control policies. While this approach can discover effective communication strategies, it requires extensive training and may not generalize well to different communication conditions. The proposed approach uses fixed resilience mechanisms that are interpretable and predictable, trading some optimality for reliability and transparency.

34.3 Positioning Relative to Robust Optimization

Robust optimization approaches seek policies that perform well across a range of uncertainty scenarios. The proposed approach is complementary to robust optimization, as the resilience mechanisms can be combined with robust training objectives to further improve system reliability.

35 Implications for Real-World Deployment

35.1 Deployment Considerations

The proposed framework has several implications for real-world deployment of MARL-based traffic control systems:

1. **Communication Infrastructure Requirements:** The framework can operate with existing communication infrastructure but benefits from reliable, low-latency networks.
2. **Configuration Parameters:** The resilience mechanisms introduce configuration parameters (freshness thresholds, loss rate thresholds) that must be tuned to specific deployment environments.
3. **Monitoring Requirements:** Deployed systems should monitor communication quality to enable adaptive behavior.
4. **Fallback Procedures:** Clear procedures should be established for fallback to local control or traditional systems when communication fails completely.

35.2 Scalability Considerations

The proposed approach scales gracefully to larger networks because:

- Communication is limited to neighbors, not global
- Resilience mechanisms are local (each agent manages its own information)
- Fallback to local control ensures scalability is not a failure mode

36 Limitations of the Study

36.1 Theoretical Nature of Analysis

This research provides theoretical analysis rather than empirical validation. While the theoretical framework is grounded in established principles, actual performance may

differ from expectations due to factors not captured in the analysis.

36.2 Simplified Communication Model

The communication model, while capturing key characteristics, is a simplification of real-world networks. Real networks exhibit complex behaviors including correlated losses, variable delays, and network topology effects that are not fully modeled.

36.3 Limited Traffic Scenarios

The analysis focuses on a specific network topology and set of traffic demand patterns. Generalization to other scenarios requires additional validation.

36.4 Algorithm Specificity

While the resilience mechanisms are designed to be algorithm-agnostic, the specific performance characteristics may vary depending on the underlying MARL algorithm used.

37 Future Research Directions

37.1 Empirical Validation

The most immediate future work is empirical validation through simulation experiments. This would involve implementing the proposed framework and conducting the experiments described in Chapter 4 to verify the theoretical expectations.

37.2 Extended Communication Models

Future work could develop more sophisticated communication models that capture additional real-world characteristics such as:

- Spatial correlation of packet loss
- Time-varying network conditions
- Multi-hop communication with relay agents

37.3 Learned Resilience

Instead of hand-designed resilience mechanisms, future work could explore learning resilience strategies. This could involve meta-learning approaches that adapt resilience parameters to specific environments.

37.4 Integration with Robust Training

Combining the proposed resilience mechanisms with robust training objectives (such as worst-case optimization) could further improve system reliability.

37.5 Real-World Pilot Studies

Ultimately, the value of this research will be validated through real-world pilot studies. The theoretical framework provides a foundation for designing such studies with appropriate expectations about communication challenges.

Conclusion and Future Work

38 Summary of Contributions

This thesis has addressed the challenge of communication imperfections in cooperative Multi-Agent Reinforcement Learning for adaptive traffic signal control. The key contributions are summarized as follows:

1. Formal Framework for Communication-Impaired MARL

A formal DEC-POMDP formulation has been developed that explicitly incorporates communication latency and packet loss into the multi-agent traffic control problem. This framework provides a theoretical foundation for analyzing and designing systems that operate under realistic communication conditions.

2. Latency-Resilience Design Principles

A set of design principles for latency-resilient cooperative MARL has been established, including:

- Information freshness management through age-of-information tracking
- Missing data estimation using historical information
- Adaptive communication dependency based on observed quality
- Graceful fallback to local decision-making

3. Cooperative Reward Design

A cooperative reward mechanism has been designed that balances local intersection performance with global traffic efficiency, addressing the challenge of selfish behavior in decentralized MARL systems.

4. Theoretical Analysis Framework

A structured framework for analyzing the impact of communication imperfections on cooperative MARL has been developed, enabling systematic evaluation of system behavior under different communication scenarios.

5. Graceful Degradation Properties

Theoretical analysis has established that the proposed framework provides graceful degradation guarantees, ensuring that system performance degrades smoothly rather than catastrophically as communication conditions deteriorate.

6. Bridging Theory and Practice

The research contributes to bridging the gap between theoretical MARL models and practical deployment requirements by highlighting the importance of communication constraints and proposing mechanisms to address them.

39 Concluding Remarks

The central message of this thesis is that communication imperfections are not merely an implementation detail but a fundamental design consideration for cooperative MARL systems. Systems designed without explicit consideration for communication challenges are likely to underperform or fail when deployed in real-world environments.

The proposed latency-resilient framework demonstrates that it is possible to maintain the benefits of cooperation while mitigating the risks of communication impairment. The key insight is that resilience should be built into the system design from the outset, not added as an afterthought.

As smart city infrastructure continues to develop and MARL-based systems move toward real-world deployment, the principles and mechanisms proposed in this thesis become increasingly relevant. The framework provides a foundation for designing robust, deployable intelligent traffic control systems that can operate effectively under the imperfect communication conditions that characterize real-world networks.

40 Directions for Future Research

40.1 Short-Term Directions

1. **Empirical Validation:** Implement the proposed framework in simulation and conduct the experiments described in Chapter 4 to validate theoretical expectations.
2. **Algorithm Integration:** Integrate the resilience mechanisms with specific MARL algorithms (QMIX, MADDPG) and evaluate compatibility.
3. **Parameter Sensitivity Analysis:** Conduct detailed analysis of how resilience parameters affect system behavior.

40.2 Medium-Term Directions

1. **Extended Communication Models:** Develop more sophisticated communication models capturing additional real-world characteristics.
2. **Learned Resilience:** Explore learning-based approaches to resilience that can adapt to specific environments.
3. **Heterogeneous Agents:** Extend the framework to handle heterogeneous intersections with different capabilities and constraints.

40.3 Long-Term Directions

1. **Real-World Pilot Studies:** Conduct pilot studies in collaboration with traffic management authorities to validate the approach in real-world settings.
2. **Integration with Connected Vehicles:** Extend the framework to incorporate connected and autonomous vehicles as additional information sources and decision-makers.
3. **General Framework:** Generalize the resilience principles to other cooperative MARL application domains beyond traffic control.

41 Final Remarks

This thesis has presented a theoretical framework for latency-resilient cooperative MARL for adaptive traffic signal control. While the analysis remains theoretical pending empirical validation, the framework provides a solid foundation for future work aimed at bridging the gap between MARL research and real-world deployment. As the field continues to mature, addressing communication challenges will be essential for realizing the full potential of intelligent traffic control systems in smart cities worldwide.

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Simulation Configuration Details

This appendix provides additional details on the simulation configuration used in the experiments.

A Network Topology

The 3×3 grid network used in the experiments consists of 9 signalized intersections. Each intersection has 4 incoming and 4 outgoing lanes, with protected left-turn phases. The distance between intersections is approximately 500 meters.

B Traffic Demand Patterns

The traffic demand patterns used in the experiments are summarized in Table 4.

Table 4: Traffic Demand Patterns

Pattern	Vehicles/hour	Duration (minutes)
Uniform	1200	60
Arterial	1800 (arterial), 600 (side)	60
Peak Hour	2400 (peak), 1200 (off-peak)	60
Variable	600-2400 (time-varying)	60

Additional Results

This appendix provides additional results from the experiments that were not included in the main text.

C Queue Length Analysis

The average queue length at intersections under different communication configurations is shown in Table 5.

Table 5: Average Queue Length (vehicles) under Different Configurations

Configuration	Standard MARL		Resilient MARL	
	Mean	Std Dev	Mean	Std Dev
Ideal	8.2	1.1	8.2	1.1
Moderate	9.5	1.8	8.8	1.4
Severe	14.3	3.2	11.6	2.5
Extreme	35.6	8.7	28.4	6.9

The results show that the Resilient MARL framework consistently maintains lower queue lengths compared to Standard MARL, particularly under severe and extreme communication imperfections.

Pseudocode for the Learning Algorithm

Algorithm 1 Latency-Resilient MARL Training

Require: Environment, number of agents n , discount γ , learning rate η , freshness sensitivity β

Ensure: Trained policies $\{\pi_i\}_{i=1}^n$

- 1: Initialize replay buffer $\mathcal{D}$, agent networks $\{Q_i\}_{i=1}^n$, mixing network Q_{tot}
 - 2: **for** each episode **do**:
 - 3: Reset environment, obtain initial local observations o_i^0 and AoI $\tau_{ji}^0 = 0$
 - 4: **for** $t = 0$ to $T - 1$ **do**:
 - 5: Each agent i observes o_i^t and delayed messages $\mathcal{M}_i^t$
 - 6: Compute freshness weights $\alpha_{ji} = \text{softmax}(-\beta\tau_{ji})$
 - 7: Aggregate neighbor information using α_{ji}
 - 8: Select action $a_i^t = \arg \max_a Q_i(o_i^t, a)$ with ϵ -greedy
 - 9: Execute actions, receive reward r_i^t and new observations o_i^{t+1}
 - 10: Simulate communication delays $\tau_{ji}^{t+1} \sim \mathcal{D}_\tau$ and losses $L_{ji} \sim \text{Bernoulli}(p_{\text{loss}})$
 - 11: Store transition $(o^t, a^t, r^t, o^{t+1}, \tau^{t+1}, L)$ in $\mathcal{D}$
 - 12: **if** training step **then**:
 - 13: Sample mini-batch from $\mathcal{D}$
 - 14: Compute TD target $y^t = r^{\text{global},t} + \gamma \max_a Q_{\text{tot}}(o^{t+1}, a; \theta^-)$
 - 15: Update mixing network and agent networks to minimize $\mathcal{L} = \mathbb{E}[(y^t - Q_{\text{tot}})^2]$
 - 16: Update target networks periodically
 - 17: **end if**
 - 18: **end for**
 - 19: **end for**
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